



OPEN SS-EMERGE - self-supervised enhancement for multidimension emotion recognition using GNNs for EEG

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Self-supervised learning (SSL) is a potent method for leveraging unlabelled data. Nonetheless, EEG signals, characterised by their low signal-to-noise ratio and high-frequency attributes, often do not surpass fully-supervised techniques in cross-subject tasks such as Emotion Recognition. Therefore, this study introduces a hybrid SSL framework: Self-Supervised Enhancement for Multidimension Emotion Recognition using Graph Neural Networks (SS-EMERGE). This model enhances cross-subject EEG-based emotion recognition by incorporating Causal Convolutions for temporal feature extraction, Graph Attention Transformers (GAT) for spatial modelling, and Spectral Embedding for spectral domain analysis. The approach utilises meiosis-based contrastive learning for pretraining, followed by fine-tuning with minimal labelled data, thereby enriching dataset diversity and specificity. Evaluations on the widely-used Emotion recognition datasets, SEED and SEED-IV, reveal that SS-EMERGE achieves impressive Leave-One-Subject-Out (LOSO) accuracies of 92.35% and 81.51%, respectively. It also proposes a foundation model pre-trained on combined SEED and SEED-IV datasets, demonstrating performance comparable to individual models. These results emphasise the potential of SS-EMERGE in advancing EEG-based emotion recognition with high accuracy and minimal labelled data.

Keywords EEG, Self-Supervised Learning, GNN, Constrastive Learning, Cross-Subject Emotion Recognition

The rapid advancement in data acquisition technologies and the exponential growth in data volume have made it increasingly inadequate to rely solely on supervised datasets to gain insights into environmental phenomena or understand the human brain's complexities. Supervised learning paradigms require vast amounts of labelled data, often expensive, time-consuming, and limited scope. In fields such as Computer Vision (CV)^{1,2}, Natural Language Processing (NLP)^{3,4}, and Audio Processing^{5,6}, recent progress has underscored the significance of leveraging extensive data available on the internet to develop foundation models. These models capture fundamental principles governing these modalities or their intermodal relationships, enabling robust performance even with limited labelled data.

Understanding brain dynamics through Electroencephalogram (EEG) signals provides crucial insights into neurological activities. These non-invasive signals are crucial in applications like emotion recognition, stress estimation, and sleep staging^{7–9}. In the context of EEG-based emotion recognition, evaluations can be conducted in several ways to validate the robustness and generalizability of models:

1. **Subject-Independent Evaluation (a.k.a. Cross-Subject or LOSO):** This method tests the model's ability to generalise across individuals. It involves training the model on data from a subset of subjects and testing it on data from entirely new subjects not included in the training set. This evaluation is crucial for applications where the model must perform well on new users without retraining.
2. **Subject-Dependent Evaluation:** This approach tests the model on the same subjects but on different sessions or instances than those used for training. It assesses how well the model performs when it has already been exposed to data from the test subjects during training. This type of evaluation might be less challenging than subject-independent tests since the model is already tuned to the characteristics of the specific subjects.

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3. **Subject-Mixed Evaluation (a.k.a. Trial-Level Random Split):** In this evaluation, all EEG trials across subjects are pooled together and randomly split into training and testing sets (e.g., 80-20), without ensuring subject separation. As a result, trials from the same individual may appear in both training and test sets. While this setup facilitates rapid benchmarking of model architectures, it does not accurately reflect cross-subject generalization and may lead to inflated performance due to subject-specific feature leakage.

These evaluation strategies are essential for developing EEG-based systems that are both effective and adaptable to various real-world scenarios. However, unlike more straightforward data types encountered in Computer Vision (CV) or Natural Language Processing (NLP), EEG data collection is fraught with unique challenges. These challenges include variability in sensor types, data collection errors, and individual differences in demographics and health, all of which contribute to high noise levels and low signal. These issues are particularly pronounced in emotion recognition, where the relevant signals can be subtle and frequently masked by artifacts or unrelated physiological events. Representation learning isolates and enhances the informative components of EEG signals crucial for distinguishing between emotional states. It mitigates the need for extensive labelled datasets for each emotional state by leveraging the intrinsic properties of the data.

Self-supervised learning (SSL), a prominent method within representation learning, excels in extracting rich representations from unlabeled EEG data^{10–13}. By learning from the inherent variability among individuals, SSL facilitates the development of models that are accurate and capable of generalising across diverse subjects^{14,15}. Once trained, these models can be fine-tuned for specific downstream tasks, such as emotion classification, with only minimal labelled data. It represents a significant paradigm shift in machine learning, especially for applications with scarce or expensive labelled data. The SSL techniques can be broadly categorised into two main types:

- *Generative Models:* These models aim to reconstruct their input data, thereby learning rich and detailed data representations. The process involves predicting missing parts of data or generating new data instances based on the observed data, which helps the model capture the underlying structure of the input distribution.
- *Contrastive Models:* These models learn by comparing and contrasting different data instances. They typically work by distinguishing between similar (positive) and dissimilar (negative) pairs of data instances, which sharpens their ability to discern critical features that define the data.

Both approaches have proven effective across various domains, including image processing, natural language understanding, and audio recognition, by leveraging large amounts of unlabeled data to learn valuable representations without explicit annotations. Specifically, in the context of EEG signals, a survey by Rafiei et al.¹⁶ indicates that generative modelling-based SSL techniques excel with smaller datasets. In comparison, contrastive learning techniques benefit from larger datasets.

Recent advancements in Self-Supervised Learning (SSL) have significantly impacted emotion recognition using EEG signals^{17–20}. Based on state-of-the-art research, the following summary illustrates diverse SSL techniques and their applications to EEG data analysis, as detailed in Table 1.

Background and related work

The Multi-view Spectral-Spatial-Temporal Masked Autoencoder (MV-SSTMA)¹⁷ employs a generative SSL approach to reconstruct signals across spectral, spatial, and temporal domains using the Masked AutoEncoder (MAE) technique, achieving high accuracy on the SEED dataset²¹. However, it shows a significant decrease in performance in self-supervised settings compared to fully-supervised ones. Additionally, MV-SSTMA's¹⁷ reliance on multi-head attention for spatial modelling assumes all channels are interconnected, potentially leading to overfitting, as it does not consider the actual electrode placements.

Graph-based Multi-Space Self-Supervised (GMSS)¹⁹ utilises a spatial contrastive approach via Simple Contrastive Learning (SimCLR)¹¹ within a Graph Neural Network (GNN). It models spatial and spectral domains, showing decent cross-subject performance on SEED²¹, but lacks robustness on the more challenging SEED-IV²² dataset, indicating limitations in model adaptability across datasets.

Work	SSL Tech.	Model	Domains	SEED ²¹		SEED-IV ²²		Limitations
				Fully SL	SSL	Fully SL	SSL	
MV-SSTMA ¹⁷	Generative	Multi-Domain reconstruction (MAE)	Spectral, Spatial and Temporal	95.32	83.49	92.82	82.55	Doesn't explore the Contrastive Learning
GMSS ¹⁹	Spatial Contrastive (SimCLR)	GNN	Spatial and Spectral	86.52	76.04	86.67	65.31	Cross-subject is decent across SEED ²¹ but poor on SEED-IV ²² shows it lacks across different datasets.
DS-AGC ²⁰	Contrastive (SimCLR)	GNN	Spatial and Spectral	85.00	82.18	65.79	61.32	SEED ²¹ numbers are good for cross-subject whereas SEED-IV ²² fully-supervised numbers show that the model is not robust.
SGMC ¹⁸	Meiosis-based Contrastive Learning	ResNet(CNN)	Temporal	94.04	89.83	NA	NA	Only models in the temporal domain, without fully exploiting the potential of spatial interactions between channels. Also, it does not evaluate cross-subject evaluation.

Table 1. State of the art Emotion Recognition work done using SSL. SL: Supervised Learning, SSL: Self Supervised Learning, Tech.: Technique.

Differential Entropy-based Self-Attentive Graph Convolution (DS-AGC)²⁰, which incorporates spatial contrastive learning with Simple Contrastive Learning (SimCLR)¹¹ and a Graph Neural Network (GNN) architecture, demonstrates good cross-subject performance on the SEED dataset²¹. However, it struggles with robustness on the SEED-IV dataset²² in both fully-supervised and self-supervised settings. It indicates that the model architecture may not be sufficiently adaptable or resilient, as it fails to perform effectively even in fully-supervised scenarios.

Self-Supervised Group Meiosis Contrastive Learning (SGMC)¹⁸ implements a contrastive model using SimCLR¹¹, over a group of subjects coming from the same stimuli while performing meiosis augmentation and contrastive loss for self-supervised learning with a Convolutional Neural Network (CNN) architecture, focusing on the temporal domain. It shows outstanding results on SEED²¹ but does not evaluate its performance on SEED-IV²², nor fully utilises spatial interactions between EEG channels. Furthermore, it lacks evaluation across cross-subject variations. Enhancing contrastive learning with meiosis-based techniques improves model performance by creating more diverse negative samples.

The works summarised in Table 1 collectively highlight a recurrent theme: while SSL techniques have significantly advanced emotion recognition from EEG data, they still face challenges in maintaining robustness across diverse subjects and datasets. These limitations often stem from model architectures that focus predominantly on individual domains such as temporal, spatial, or spectral—without effectively integrating them.

Causal Convolutions, as employed in Temporal Convolutional Networks (TCNs), address temporal dependencies by ensuring that each time step is influenced only by the current and past inputs, preserving the causality intrinsic to EEG signals^{23–25}. This feature is particularly beneficial for emotion recognition, where the sequence of brain activity reflects emotional transitions. Furthermore, dilated convolutions allow TCNs to capture long-range temporal dependencies without excessive computational overhead, providing an efficient alternative to recurrent networks.

Graph Attention Networks (GATs) dynamically enhance spatial domain modelling by assigning varying importance to interactions between EEG channels. Unlike static graph methods, GATs utilise an attention mechanism that identifies the most relevant electrode relationships for a given emotional or cognitive state, making them highly adaptable to the variability present in EEG data^{26,27}. This capability is crucial for capturing spatial dependencies, as interactions between brain regions vary across subjects and tasks. GATs prioritise key spatial patterns, improving robustness in subject-dependent scenarios.

Spectral Embeddings leverage frequency-domain features to encode EEG signals into a compact and meaningful vector space. Differential Entropy (DE), in particular, has been widely recognised as one of the most practical features for emotion recognition, as it captures variability across critical frequency bands (e.g., delta, theta, alpha, beta, and gamma)^{7,17}. By deriving DE-based embeddings before modelling spatial and temporal aspects, SS-EMERGE ensures that frequency-domain information is preserved and seamlessly integrated into the overall representation.

The combination of these techniques—temporal modelling with Causal Convolutions, spatial relationship modelling with GATs, and frequency-domain encoding with Spectral Embeddings—offers a unified and holistic framework in SS-EMERGE. Each component contributes uniquely to addressing the inherent challenges of EEG-based emotion recognition. Temporal modelling captures the dynamic evolution of emotional states, spatial modelling identifies critical inter-channel relationships, and spectral embeddings provide robust frequency-domain features. Together, these techniques enable SS-EMERGE to achieve superior generalizability and performance across diverse datasets and subject variations, overcoming the limitations of methods that prioritise single-domain representations.

This work proposes a comprehensive approach to enhance EEG-based emotion recognition by integrating self-supervised learning with multidimension graph neural networks. The key innovations and contributions of this work are as follows:

1. Proposal of SS-EMERGE model for SSL training for Emotion Recognition with main contributions as follows:
 - (a) *Temporal, Spatial, and Spectral Integration*: Combines causal convolutions for encoding temporal interactions, and using Graph Attention Transformer (GAT)²⁸ inspired by famous multivariate time series anomaly detection model, MTAD-GAT²⁷ for learning spatial channel interactions, and borrowing integration of spectral domain by using spectral embeddings from MV-SSTMA¹⁷. The multidimension integration ensures a holistic representation of EEG signals, capturing the intricate dynamics in different domains.
 - (b) *Meiosis Augmentations Based Contrastive Learning*: Integration of the meiosis-based contrastive learning technique proposed by SGMC¹⁸ for EEG. It performs contrastive learning on groups of stimuli, such as video samples in SEED²¹ and SEED-IV²², instead of individual subject samples. The augmentation technique cross-exchanges EEG segments within a video group, creating new augmented samples that facilitate contrastive learning and enhance the model's ability to generalise across different subjects and emotional states.
 - (c) *Foundation Model for SEED and SEED-IV*: Proposes a foundation model for both SEED²¹ and SEED-IV²² datasets that can be further fine-tuned on data collected through the same EEG sensor used by SEED²¹ and SEED-IV²² to be used for the downstream task of Emotion Classification.
2. Evaluates Subject Dependent performance on SEED²¹ and SEED-IV²² datasets:

- (a) Subject dependent SSL performance with respect to fully-supervised learning for SGM¹⁸ and SS-EMERGE.
- (b) The effect of foundational pretraining while combining SEED²¹ and SEED-IV²² on cross-subject classification performance.

This work introduces a novel SS-EMERGE model that integrates SSL and GNN for advanced emotion recognition from EEG data. It addresses key challenges, including the need for extensive labelled datasets and handling high noise and variability in EEG signals. The SS-EMERGE model achieves 95.05% accuracy on SEED²¹ and 83.67% on SEED-IV²² with minimal labelled data for emotion recognition.

The structure of this paper is organised as follows: First, Section 2 defines the problem statement and details the proposed solution. Next, Section 4 showcases the experimental results obtained by applying the methodology described in Section 2. It is followed by a Section on Experimental Setup. Finally, Section 5 summarises the findings and discusses potential future research directions.

Methodology

This section details SS-EMERGE for emotion recognition using a comprehensive framework that integrates multiple domains of EEG data processing. The methodology includes a pretraining phase with Meiosis-based contrastive learning followed by a fine-tuning phase with labelled data. The system processes EEG data in spectral, spatial, and temporal domains, utilising specialised network components for each aspect.

Figure 1 describes the overall flow of SS-EMERGE that integrates a dual-phase approach to enhance the effectiveness of an EEG-based emotion recognition model. Phase-1 (pretraining) and Phase-2 (fine-tuning) followed by evaluation for the held-out subject, are briefly explained as follows:

1. **Phase-1 (pretraining without labels):** This phase learns the generalised features from a diverse EEG dataset using self-supervised contrastive learning. This phase employs components such as meiosis-based contrastive learning without using the emotion labels of the sample. By leveraging data from all subjects except one (N-1 subjects), this phase captures broad patterns and features inherent in the entire dataset, promoting generalisation across different contexts and conditions²⁹.
2. **Phase-2 (Fine-tuning with labels):** The fine-tuning phase aims to adapt the pre-trained model to the downstream emotion classification task. This phase involves fine-tuning the pre-trained model with a classifier head using Cross-Entropy Loss on the same N-1 subjects. It ensures that the model is specifically tailored to recognise the emotional states from the EEG data of the given subjects.
3. **Prediction on the hold-out subject:** Finally, the prediction happens on all the sessions of the hold-out subject (the Nth subject) for evaluation.

Model architecture

The SS-EMERGE model architecture leverages a combination of advanced techniques to process EEG signals for emotion recognition effectively. This section explains each component in the model architecture, from input and initial processing, spectral, temporal, and spatial modelling to latent representation modelling. Figure 2

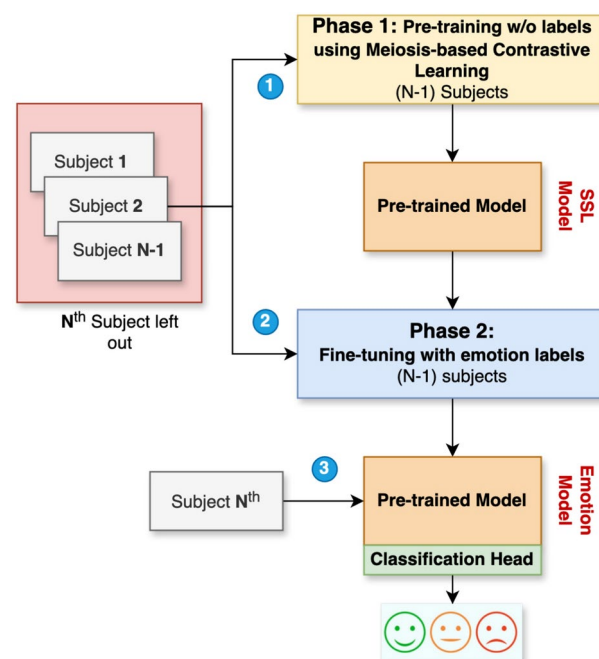


Fig. 1. SS-EMERGE Flow Diagram.

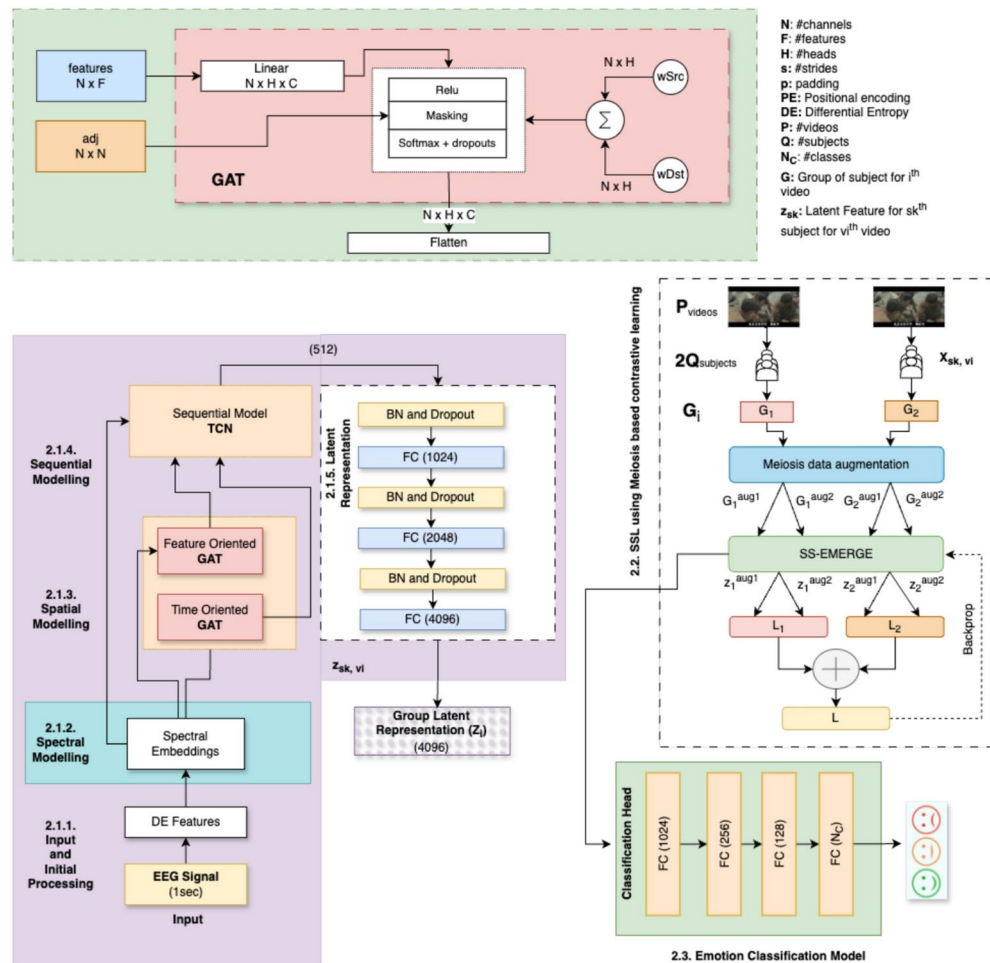


Fig. 2. SS-EMERGE backbone architecture for representation learning using SSL and fine-tuning for emotion classification label and associate with the text not clear.

illustrates the SS-EMERGE backbone architecture that learns the latent representations in Phase-1 (pretraining), followed by Phase-2 (fine-tuning), which fine-tunes the pre-trained network adding the classification for downstream emotion recognition task.

Input and initial processing

The input to the model consists of raw EEG signals segmented into 1-second intervals. These short segments help capture the immediate brain responses associated with different emotional states. From these EEG signals, Differential Entropy (DE)^{9,30} features are extracted. DE features are robust in representing the spectral properties of EEG signals, making them particularly suitable for emotion recognition tasks.

Spectral domain modelling

SS-EMERGE leverages the Differential Entropy (DE) feature, which has demonstrated effectiveness in EEG-based emotion recognition tasks³¹. The model extracts DE features $X = (x_1, x_2, \dots, x_N) \in \mathbb{R}^{N \times C \times F}$ from EEG signals in the spectral domain and transforms them into samples $\tilde{X} = (\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_N) \in \mathbb{R}^{N \times C \times T \times F}$ using an overlapping window of T seconds, where F represents frequency bands (δ : 1–4 Hz, θ : 4–8 Hz, α : 8–14 Hz, β : 14–31 Hz, and γ : 31–50 Hz) converted by a Short-Time Fourier transform (STFT) for each channel C . In the spectral embedding layer, SS-EMERGE projects the i -th sample, $\tilde{x}_i$, which resides in $\mathbb{R}^{C \times T \times F}$, into a D -dimensional space using a linear layer. This process embeds the spectral information of the EEG signals into $\tilde{x}_i^{(s)} = \tilde{x}_i w^{(s)} + b^{(s)}$, where $w^{(s)}$ is the weight vector in $\mathbb{R}^{F \times D}$ and $b^{(s)}$ is the bias in $\mathbb{R}^D$. It finally reshapes $\tilde{x}_i$ to $C \times T \times D$ dimensional embedding shape. Later, Positional Encoding (PE)³² layer retains the temporal position of all channels C by adding sine-cosine encoding on each i th time step of signal with length T of every channel to facilitate later encoding.

Spatial domain modelling

Inspired by the Multivariate Time-series Anomaly Detection via Graph Attention Network (MTAD-GAT)²⁷, which first passes the time series through 1D Conv followed by two GAT networks. SS-EMERGE integrates two

types of Graph Attention Transformers (GAT)²⁶ to effectively capture spatial using Feature-Oriented GAT and temporal dependencies via Time Oriented GAT in the EEG signals.

1. **Feature Oriented GAT:** This layer constructs a graph where nodes represent different features simultaneously, and edges depict the relationships or interactions between these features. It effectively models the spatial relationships among various EEG features, enhancing the model's ability to understand how different features interact at a given moment in time.
2. **Time Oriented GAT:** This layer constructs a graph where nodes represent the same feature across different time points, and edges represent the temporal dependencies between these nodes. It captures the temporal progression of individual EEG features over time, allowing the model to track changes and developments in feature values across the duration of the signal³³.

Temporal domain modelling

In SS-EMERGE, **sequential modelling** is handled using a **Temporal Convolutional Network (TCN)** to capture temporal dependencies in EEG signals. Unlike recurrent architectures, TCN employs **causal convolutions**³⁴, ensuring that each output at a given time step is influenced only by current and past inputs while preserving the chronological order of EEG sequences.

The **TCN module** processes **spectrally and spatially enriched embeddings** using stacked **1D convolutional layers** with dilation, allowing the network to model **long-range dependencies** without the vanishing gradient problem of recurrent models. This makes it highly efficient for EEG-based emotion recognition.

As illustrated in Figure 2, sequential modelling in SS-EMERGE follows a structured pipeline:

1. **Input Features:** Spectral embeddings from the **Spectral modelling** module.
2. **Feature-Oriented and Time-Oriented GAT:** Extracts spatial relationships in EEG signals.
3. **Temporal Convolutional Network (TCN):** Models sequential dependencies over time.
4. **Batch Normalisation and Dropout Layers:** Prevent overfitting and stabilise learning.

Latent representation

Graph Attention Transformers (GATs) first process outputs through a series of Batch Normalization (BN) and Dropout Layers with a dropout rate of 0.2 before passing them to the Fully Connected (FC) layers. It starts with batch normalisation to stabilise the learning process and incorporates a dropout of 0.2 to prevent overfitting. Subsequent layers increase in neuron count from 1024 to 2048 and then 4096, each refining the features into a high-dimensional latent representation that encapsulates comprehensive EEG signal information.

Phase-1: SSL using Meiosis-based contrastive learning

Phase 1 (pretraining) learns the latent representation using Meiosis-based contrastive learning to the discriminative capabilities of a model by leveraging the augmented pairs created through Meiosis Data Augmentation. In the context of EEG data, contrastive learning learns distinct differences in brain responses that can signify various cognitive or emotional states in response to identical stimuli. Meiosis augmentation enhances EEG-based emotion recognition by cross-exchanging signal segments from different subjects within the same group. It creates augmented pairs, where positive samples come from subjects who watched the exact video (same stimuli), and negative samples come from different video groups (different stimuli). Meiosis-based Contrastive Learning helps the model to distinguish between similar and dissimilar stimuli, improving its ability to recognise emotions.

Meiosis data augmentation

Meiosis data augmentation aims to enhance EEG-based emotion recognition by augmenting one group sample to generate two groups that preserve the same stimuli-related features. It involves aligning stimuli in the EEG group to construct positive pairs, thereby increasing the meaningful difficulty of the model decoding EEG samples.

1. Group Creation:

- (a) **Sampling EEG Data:** Select $2Q$ subjects EEG recordings $(X_{s_1}, X_{s_2}, \dots, X_{s_{2Q}})$ based on their exposure to specific video clips P videos $(v_1, v_2, \dots, v_P)$, where each clip serves as a unique stimulus.
- (b) **Group Formation:** For each selected video clip v_i , form groups G_i consisting of EEG samples from the $2Q$ subjects. Thus, a group G_i for video clip v_i is defined as:

$$G_i = \{X_{s_1}^{v_i}, X_{s_2}^{v_i}, \dots, X_{s_{2Q}}^{v_i}\}$$

Each group G_i shares similar stimuli-related features among its samples.

2. Meiosis Data Augmentation:

- (a) **Individual Pairing:** For an original group of EEG signals $G_i = \{X_{s_k}^{v_i} \mid k = 1, 2, \dots, 2Q\}$ (corresponding to a video clip v_i), pair the signals randomly to form Q pairs: $\{X_{s_1}^{v_i}, X_{s_{1+Q}}^{v_i}\}, \{X_{s_2}^{v_i}, X_{s_{2+Q}}^{v_i}\}, \dots, \{X_{s_Q}^{v_i}, X_{s_{2Q}}^{v_i}\}$.

- (b) *Crossover*: Meiosis receives a randomly given split position c (where $1 < c < T - 1$) to perform the crossover transformation for each pair, exchanging data of the first c sampling points between the two signals. It results in augmented pairs $\{\{\tilde{X}_{s_k}^{v_i}, \tilde{X}_{s_{k+Q}}^{v_i}\} \mid k = 1, 2, \dots, Q\}$.
- (c) *Separation*: The transformed signals are then randomly divided into two groups. Paired transformed signals are placed in different groups X and Y , which results in two homologous groups of EEG $\tilde{G}_X^i = \{\tilde{X}_{s_k}^{v_i} \mid k = 1, 2, \dots, Q\}$ and $\tilde{G}_Y^i = \{\tilde{X}_{s_{k+Q}}^{v_i} \mid k = Q + 1, Q + 2, \dots, 2Q\}$ that share similar group-level stimuli-related features.

3. Constructing the Minibatch:

- (a) Equation 1 shows that for one minibatch of P group samples G , $2P$ group samples $\tilde{G}$ are obtained through the Meiosis augmentation process.

$$\tilde{G} = \{\tilde{G}_t^i \mid i = 1, 2, \dots, P; t \in \{1, 2\}\} = \text{Meiosis}(G) \quad (1)$$

- (b) $\tilde{G}_X^i$ forms a positive pair with $\tilde{G}_Y^i$ and forms negative pairs with any other $2(P - 1)$ group samples.

Contrastive learning setup

Contrastive learning uses meiosis-augmented pairs to train a model that learns to distinguish between pairs that share the same stimuli label (positive pairs) and pairs that come from different stimuli labels (negative pairs). Figure 2 shows that the pretraining phase maximises the similarity between positive pairs and minimises the similarity between the negative pairs in the latent space⁵. The complete contrastive learning happens along with the Meiosis Augmentation as follows:

1. *Augmentation*: Meiosis augmentation generates pairs of EEG signals $\tilde{G}_X^i$ and $\tilde{G}_Y^i$ by cross-exchanging segments from different subjects within the same group. For instance, $\tilde{G}_X^i$ might contain segments from subject i and $\tilde{G}_Y^i$ from subject j , both corresponding to the same stimuli.
2. *Feature Extraction*: SS-EMERGE extracts latent features z_X for the augmented pairs $\tilde{G}_X^i$ and $\tilde{G}_Y^i$, generated by meiosis augmentation³⁵.
3. *Similarity Measurement*: Equation 2 calculates the similarity between the latent representations z_X and z_Y using cosine similarity. It quantifies how close the representations of the augmented pairs are in the latent space³⁶.

$$s(z_X, z_Y) = \frac{z_X \cdot z_Y}{\|z_X\| \|z_Y\|} \quad (2)$$

4. *Contrastive Loss*: Finally, Phase-1 employs contrastive loss to train the model using SimCLR¹¹ loss function. The loss function penalises large deviations between the similarity scores and their expected values based on the temperature-scaled cross-entropy in Equation 3:

$$\ell_i^X = -\log \frac{\exp(s(z_i^X, z_j^Y)/\tau)}{\sum_{j=1, j \neq i}^P \exp(s(z_i^X, z_j^X)/\tau) + \sum_{j=1}^P \exp(s(z_i^X, z_j^Y)/\tau)} \quad (3)$$

Here, P denotes the number of groups, τ is the temperature parameter, and $[j \neq i]$ is an indicator function that equals 1 if j is not equal to i ³⁷.

Equation 4 then calculates the combined loss across all the groups in a batch of Stochastic Gradient Descent.

$$L = \frac{1}{2P} \sum_{i=1}^P (\ell_i^X + \ell_i^Y) \quad (4)$$

Algorithm 1 outlines the end-to-end Phase-1 pretraining using contrastive learning with meiosis augmentation. It starts by extracting group-level features and projecting them to latent space. It helps obtain group representations. The algorithm then minimises the distance between positive pairs and maximises amongst the negative pairs³⁸.

Phase-2: fine-tune emotion classification model

In Phase-2 (fine-tuning), the pre-trained SS-EMERGE model generates the latent representation but further fine-tunes using the classification head seen in Figure 2 for emotion classification modelling. It only learns the parameters of the classification head without changing other model parameters.

Model Architecture and Training Procedures:

1. **Classifier Head**: The classifier consists of additional fully connected layers that further process the latent representation of dimension D . These layers are followed by a softmax layer to produce the final emotion class probabilities (N_C).
- (a) *Fully Connected Layers*: The layers with 1024, 256, and 128 neurons progressively transform and reduce the dimensionality of the latent representation.

1: **Initialization:**
2: Set the number of epochs E_1
3: Initialize the model parameters
4: **for** each epoch $t = 1, 2, \dots, E_1$ **do**
5: **Video Clip Sampling:**
6: Sample P video clips per iteration until all video clips are enumerated
7: **EEG Sample Extraction:**
8: Extract $2P \times Q$ EEG samples $D = \{X_{sk}^{vi} | i = 1, 2, \dots, P; k = 1, 2, \dots, 2Q\}$
9: Pack these samples into groups $G = \{G_i | i = 1, 2, \dots, P\}$
10: **Meiosis Data Augmentation:**
11: Randomly generate a fixed split position c (where $1 < c < M - 1$)
12: Apply Meiosis crossover to each group to generate augmented samples $\tilde{G} = \{\tilde{G}_i^t | i = 1, 2, \dots, P; t \in \{X, Y\}\}$
13: **Feature Extraction and Projection:**
14: Extract group-level features
15: Project features to latent space to obtain group representations
16: **Loss Calculation:**
17: Calculate the loss L using Equation 3
18: **Model Update:** state Perform backpropagation to calculate gradients state
 Update the parameters of the model using the optimiser
19: **end for**

Algorithm 1. Pretraining with Meiosis-Based Contrastive Learning

- (b) *Class Logits (FC (N_C))*: This layer outputs N_C neurons, where N_C corresponds to the number of emotion classes.
- (c) *Softmax Layer*: Converts the logits into class probabilities using Softmax.
2. **Training Procedure:**

(a) Finally, Phase 2 fine-tunes the classifier head over the pre-trained model using labelled EEG data.

(b) The training process involves minimising a cross-entropy loss, which learns to discriminate different predicted class probabilities and the true emotion labels.

(c) It optimises the network using Stochastic Gradient Descent (SGD), with appropriate learning rate scheduling to ensure convergence shown in Table 2.

(d) It regularises the model training using a dropout of 0.5 and batch normalisation to prevent overfitting.

Phase	Parameter	Value
SSL Phase	Learning Rate	1×10^{-3}
	Momentum	0.1
	Batch Size	64
	Epochs	3288
	Meiosis Augmentation - P	16 (video clips per iteration) (SGMC ¹⁸)
	Meiosis Augmentation - Q	2 (samples per group) (SGMC ¹⁸)
Classification Phase	Learning Rate	1×10^{-3}
	Batch Size	256
	Epochs	100
	Optimization	Adam with learning rate scheduling
	Loss Function	Cross-entropy
	Dropout	0.5

Table 2. Hyperparameters of the SS-EMERGE Model.

Experimental setup

Dataset

This work conducts experiments on the SEED²¹ and SEED-IV²² datasets, known for their robust emotional state annotations and varying class distributions. These datasets provide a comprehensive and balanced foundation for evaluating the performance of emotion recognition models.

1. **SEED-IV Dataset**²¹: Consists of EEG recordings from 15 subjects (7 males and 8 females) as they watched film clips designed to evoke specific emotions. Each subject contributed 45 samples, evenly distributed among three emotion labels: happy, sad, and neutral, resulting in a total of 675 samples²¹. The signals were recorded using a 62-channel ESI NeuroScan System, with a sampling rate of 1,000 Hz. To facilitate analysis, the signals were downsampled to 200 Hz and bandpass filtered between 0–75 Hz to remove unrelated artifacts.
2. **SEED-IV Dataset**²²: Includes data from the same 15 subjects, with each providing 72 samples distributed across four emotion labels: happy, sad, fear, and neutral, leading to a total of 1,080 samples²². The experimental setup and data processing methods were similar to those used in the SEED²¹ dataset. However, the bandpass frequency filter was set to 1–75 Hz for SEED-IV, and the DE features were normalised to ensure consistency across samples.

The balanced distribution of samples across subjects and emotion classes in both datasets ensures an unbiased training and evaluation process.

Data preparation

For both SEED²¹ and SEED-IV²² datasets, each trial of EEG signal in each channel undergoes L2 normalisation to ensure that the data from different trials are on a similar scale. The movie videos are divided into 1-second windows. Given the varying lengths of the trial videos, adjacent windows are segmented from front to back according to the time axis until the coverage of windows exceeds the video range.

Evaluation strategy

To rigorously assess the generalisation capabilities and performance stability of the SS-EMERGE model, this work employs a robust evaluation strategy involving subject-mixed and subject-independent testing frameworks. The Leave-One-Subject-Out (LOSO) cross-validation scheme is utilised to ensure comprehensive evaluation:

1. Subject-Independent / Cross-Subject Evaluation (LOSO):

- **Fully-supervised Learning:** For each cycle of LOSO, the model is trained on EEG data from $N - 1$ subjects using all available labels. The model is then tested on the left-out subject, assessing its adaptability to new individual data unseen during the training phase.
- **Self-Supervised Learning:** Begins with pretraining a base model on all subjects without labels to capture generalisable features. The model is then fine-tuned on minimal labelled data from the left-out subject, optimising the blend of learned representations with subject-specific adjustments.

2. Subject-Mixed / Trial-Random Evaluation:

- Conducted using a random train-test split (80–20%) to evaluate the model's architecture and ability to generalise across a random data selection, focusing not specifically on individual subject differences.

Training parameters for SSL and classification model

Phase-1 (pretraining) and Phase-2 (fine-tuning) select the hyperparameters used by SGM¹⁸ along with the default settings of the optimiser. These settings yield good subject-independent and subject-dependent performance without the rigorous need for hyperparameter tuning. Table 2 lists all the hyperparameters with their corresponding default values during pretraining and fine-tuning of the Emotion Classification model.

Results and discussion

The results presented in this section evaluate the performance of SS-EMERGE in different experimental settings in the SEED²¹ and SEED-IV²² datasets. The evaluation focuses on three key aspects as follows:

- Experiment 1: Subject-Mixed and Cross-Subject Evaluation of Base-Model Architecture Using fully-supervised Learning
- Experiment 2: Comparative Analysis of Self-Supervised vs Full-Supervised Learning for Cross-Subject Adaptation
- Experiment 3: Foundational pretraining by combining SEED and SEED-IV.

Experiment 1: choosing a base model architecture

EEG-based emotion recognition is inherently challenging due to subject-specific variations. A cross-subject generalisation is particularly difficult, as models must adapt to significant intersubject variability. This experiment compares the ResNet used by SGM¹⁸ with the proposed SS-EMERGE, a fully-supervised model trained for EEG emotion classification. The objective is to determine the most effective fully-supervised backbone model for emotion recognition for cross-subject.

Model	Setup	Positive (%)	Neutral (%)	Negative (%)	Average (%)
ResNet	Subject-Mixed	89.7	90.0	89.8	89.83
	Cross-Subject	68.5	69.8	65.2	67.83
SS-EMERGE	Subject-Mixed	96.2	92.5	94.1	94.27
	Cross-Subject	87.4	83.1	81.9	84.13

Table 3. Performance of models on SEED²¹ dataset under subject-independent and cross-subject setups.

#Samples	SGMC	SS-EMERGE
SEED Dataset ²¹		
10% (fine-tuned)	73.29%	75.50%
50% (fine-tuned)	82.91%	86.13%
100% (fine-tuned)	90.75%	92.35%
SEED-IV Dataset ²²		
10% (fine-tuned)	52.30%	58.10%
50% (fine-tuned)	68.05%	76.75%
100% (fine-tuned)	75.20%	81.51%

Table 4. Self-Supervised Learning Comparison of SGMC¹⁸ and SS-EMERGE on SEED²¹ and SEED-IV²² datasets.

Unlike previous studies that rely primarily on subject-independent evaluations, this work evaluates cross-subject, ensuring that the model is trained on $N - 1$ samples of a partial sample of subjects and tested on rest. This approach rigorously assesses cross-subject generalisation.

From Table 3, ResNet achieves 67.83% accuracy under the cross-subject evaluation, demonstrating its limited ability to generalise across subjects. In contrast, SS-EMERGE outperforms ResNet, achieving 84.13%, highlighting its effectiveness as a fully-supervised backbone model for EEG-based emotion recognition. Both models exhibit stronger performance in the random-split setting due to similarities in intra-subject training and testing data.

Experiment 2: cross-subject evaluation under self-supervised and fully-supervised learning

Having established SS-EMERGE as the superior fully-supervised backbone in Experiment 1, this experiment evaluates SS-EMERGE, a self-supervised learning (SSL) approach that combines contrastive learning with classification. SS-EMERGE is compared against SGMC, which also employs SSL, and SS-EMERGE, the strongest fully-supervised model from Experiment 1.

The SEED²¹ and SEED-IV²² datasets are used to analyse how SSL compares with fully-supervised learning under cross-subject conditions. The model is fine-tuned using varying percentages of labelled data (10%, 50%, and 100%) to evaluate performance under different data availability scenarios.

From Table 4, SS-EMERGE consistently outperforms both SGMC and SS-EMERGE across all labelled data settings. Notably, when fine-tuned with only 10% labelled data, SS-EMERGE achieves 75.5% accuracy on SEED, compared to 74.12% for SS-EMERGE and 73.29% for SGMC. Also, when compared with fully-supervised counterpart, SEED²¹ achieves 84.13% cross-subject performance as compared to 92.35% of SSL.

Experiment 3: foundational pretraining using SEED and SEED-IV datasets

This experiment evaluates the impact of foundational pretraining using the combined datasets of SEED²¹ and SEED-IV²². Given that both datasets share identical channel configurations, data integration during pretraining is streamlined. The distinct video stimuli in each dataset require a specialized approach for stimulus identification. To address this, a composite identifier that merges the dataset and video ID is used, ensuring accurate alignment with subject-specific emotional responses.

Baseline models

The baseline models for this study include:

- **Vanilla SGMC:** This model operates solely in the temporal domain using a ResNet architecture, focusing on capturing dynamic temporal features.
- **SS-EMERGE:** An advanced model that integrates features across temporal (T), spectral (F), and spatial (S) domains using a Graph Attention Transformer (GAT)²⁸.

Ablation studies

Ablation studies are performed to assess the impact of each domain:

1. **SS-EMERGE.FT:** Focuses on the spectral and temporal domains, evaluating the models ability to utilize these features for emotion recognition.

	SGMC	SS-EMERGE.FT	SS-EMERGE.TS	SS-EMERGE
#Samples	T	FT(Proposed)	TS(Proposed)	FTS(Proposed)
SEED Dataset ²¹				
10%	73.29%	73.82%	73.51%	74.11%
50%	83.71%	84.02%	83.84%	84.42%
100%	83.04%	85.56%	85.12%	88.87%
SEED-IV Dataset ²²				
10%	58.33%	56.81%	55.53%	57.92%
50%	66.44%	62.32%	61.02%	63.54%
100%	77.64%	76.31%	75.08%	82.75%

Table 5. SSL Performance with foundational pretraining for SEED²¹ and SEED-IV²².

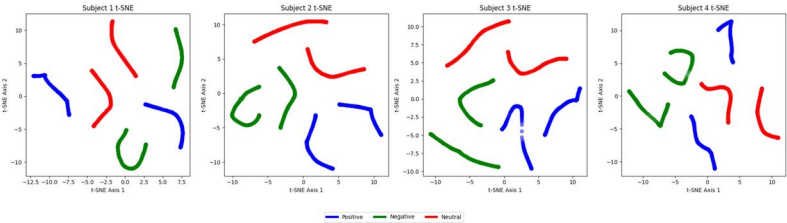


Fig. 3. TSNE visualization of SEED²¹ embeddings.

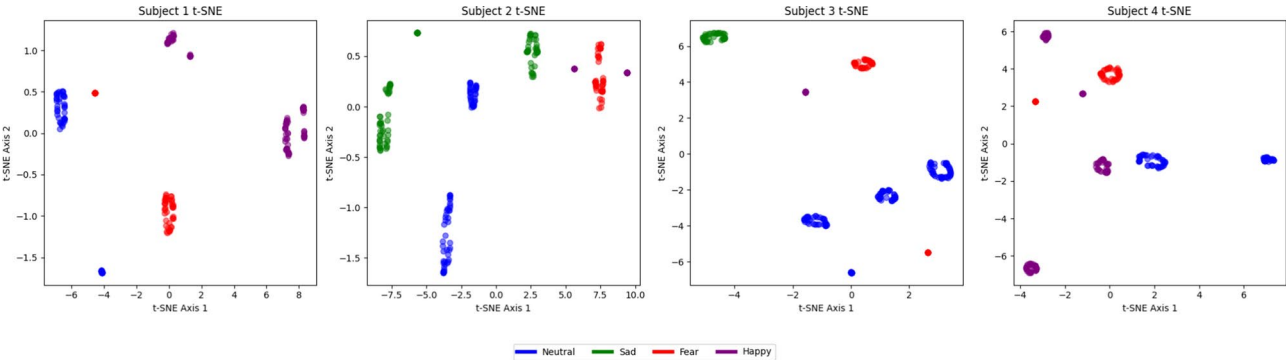


Fig. 4. TSNE visualization of SEED-IV²¹ embeddings.

2. **SS-EMERGE.TS:** Analyzes the integration of temporal and spatial domains, omitting spectral features to determine the importance of spatial information.

These models are evaluated on a subject-independent basis due to computational constraints, which also guide the domain-specific ablation studies.

The SEED²¹ and SEED-IV²² datasets are processed under identical protocols to ensure uniformity, as detailed in subsection 3.2. Stimuli are uniquely tagged with dataset-specific prefixes, facilitating clear differentiation and integration into the SGMC model without modifications. Foundational pretraining employs contrastive loss to enhance feature discrimination across emotional stimuli without using labels.

In the fine-tuning phase, labels are reintroduced, tailoring the model to the distinct emotional patterns of each dataset. Performance is evaluated directly on the subjects from the training data, avoiding cross-subject validation.

As shown in Table 5, pretraining on the combined datasets significantly improves model accuracy 88.80% on SEED and 81.50% on SEED-IV^{21,22}, demonstrating the efficacy of cross-domain pretraining. This approach represents the first attempt to merge and assess these datasets for cross-subject performance.

Experiment 4: visualisation of underlying embedding

The t-Distributed Stochastic Neighbor Embedding (t-SNE) visualisations in Figures 3 and 4 illustrate the feature representations from the SS-EMERGE on the SEED²¹ and SEED-IV²² dataset. The t-SNE projections reduce the 4096-dimensional feature space to a 2-dimensional space for visualisation. The clustering of SS-EMERGE learnt

features shows distinct and well-separated clusters. The more precise separation of clusters of the SS-EMERGE indicates that it enhances the model's ability to capture the discriminative features due to its rich representation learning.

Conclusion and future work

This study presents the SS-EMERGE model, a novel self-supervised learning approach using multidimension graph neural networks to enhance EEG-based emotion recognition. Evaluated on the SEED and SEED-IV datasets, SS-EMERGE consistently outperforms the SGMCM baseline and fully-supervised models, particularly when labelled data is limited. The model shows substantial performance gains with just 10% and 50% labelled data and achieves its highest accuracy with full data utilisation, highlighting its potential for applications with scarce labelled resources.

Furthermore, SS-EMERGE demonstrates its capability as a foundational model for emotion recognition by efficiently leveraging combined datasets during pretraining. Future research could explore integrating additional diverse EEG datasets to improve model robustness further and investigate the efficacy of meiosis-based contrastive learning for enhanced group-level feature representation. This direction is promising for advancing emotion recognition technologies in real-world scenarios.

Data availability and access

The data used in this study is available on SEED²¹ and SEED-IV²². Access to the datasets can be requested through the following link: <https://bcmi.sjtu.edu.cn/home/seed/>.

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Author contributions

The authors confirm contribution to the paper: study conception and design: Chirag Ahuja; analysis and interpretation of results: Chirag Ahuja; draft manuscript preparation: Chirag Ahuja, Divyashikha Sethia. All authors reviewed the results and approved the final version of the manuscript.

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